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Chapter 7: Assessing Habitability Beyond Earth

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Abstract

All known life on Earth inhabits environments that maintain conditions between certain extremes of temperature, chemical composition, energy availability, and so on (Chapter 6). Life may have emerged in similar environments elsewhere in the Solar System and beyond. The ongoing search for life elsewhere mainly focuses on those environments most likely to support life, now or in the past—that is, potentially habitable environments. Discussion of habitability is necessarily based on what we know about life on Earth, as it is our only example. This chapter gives an overview of the known and presumed requirements for life on Earth and discusses how these requirements can be used to assess the potential habitability of planetary bodies across the Solar System and beyond. We first consider the chemical requirements of life and potential feedback effects that the presence of life can have on habitable conditions, and then the planetary, stellar, and temporal requirements for habitability. We then review the state of knowledge on the potential habitability of bodies across the Solar System and exoplanets, with a particular focus on Mars, Venus, Europa, and Enceladus. While reviewing the case for the potential habitability of each body, we summarize the most prominent and impactful studies that have informed the perspective on where habitable environments are likely to be found. Key Words: Habitability—Habitable worlds—Icy moons—Mars—Venus—Europa—Enceladus. Astrobiology 24, S-143–S-163.

The search for life beyond Earth begins with an assessment of whether a given planetary body is potentially habitable. For an environment to be considered habitable, it must have certain planetary characteristics (Chapter 3.4) and host environments (Chapter 4.3) that support life as we know it. Although it is conceivable that life elsewhere may exist under conditions more extreme than where we find life on Earth, we have no evidence that such life exists. Thus, it is customary to use the limits of life on Earth (Chapter 6.3) as a guide to interpret the potential habitability of environments elsewhere. This approach helps us determine where to search for life elsewhere and results in testable hypotheses of potential habitability that may be evaluated through biosignatures (Chapter 8).

Chapter 7.1 discusses the concept of habitability and how the study of astrobiology informs the assessment of potentially habitable environments. Then, Chapter 7.2 details current consensus on the potential habitability of various environments in the Solar System, and considerations for exoplanets that may be habitable. Assessing habitability is not possible without synthesis of concepts found in all previous chapters in this primer, highlighting the interrelated nature of astrobiology research.

7.1. Requirements for Habitable Environments

Identifying favorable environments for life depends on what we consider ''life.'' A strict definition for life is notoriously contentious, as discussed in Chapter 2.4. How life began on Earth is unknown, yet the conditions relevant to possible origins are critical to understanding where habitable environments may be found. Our understanding of what constitutes a habitable environment on Earth has also changed over time and with better understanding of extremophiles. Based on modern ideas about life and habitability on Earth, in this chapter we consider how that understanding is applied to other planetary bodies. Factors affecting habitability are summarized in Fig. 7.1.

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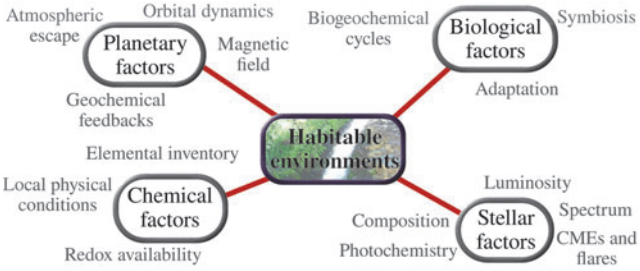


FIG. 7.1. Overview of the many interrelated factors and processes that affect considerations of habitability. Many factors may be grouped into multiple categories. CMEs = coronal mass ejections.

Potentially habitable regions can be predicted based on the availability of basic requirements, such as the essential ''CHONPS'' elements (Chapter 2.2.2), mild temperatures and pressures, and limited ionizing radiation. In practice, it is much simpler to determine which environments are not habitable, because an environment outside the livable range of even one essential characteristic renders the system uninhabitable. For example, the surface of Venus is much hotter than known life can tolerate and nearly devoid of water (Taylor *et al.,* 2018). All life on Earth requires liquid water to grow and reproduce (Chapter 2.2.2.1). Therefore, it is reasonable to conclude that the hot and dry surface of Venus cannot currently support life as we know it. Conversely, it is difficult to verify whether an environment contains all life-essential ingredients. There are many potentially habitable environments in the Solar System (and beyond) that may have the requirements for life to survive, either now or in the past. Potentially habitable environments are discussed in more detail in Chapter 7.2.

It is important to note that discussion of habitability does not mean that the environment may support humans or other familiar forms of complex life. Most potentially habitable environments on other astronomical bodies are similar to environments that we would consider extreme on Earth (such extreme Earth environments are discussed in Chapter 6.3). Extremophiles live at the boundaries of where known life can exist, and they are all microscopic (Pikuta *et al.,* 2007). For the majority of Earth's history, the only life-forms were single-celled organisms (see Chapter 5.4.2). This suggests that if we were able to take a snapshot of extant life in a habitable environment at some random point in the life-filled history of a planetary object, the observed life would most likely be microbial. It is possible that environments that have been habitable for far longer than Earth may never develop complex life with multicellular organisms. We cannot know with certainty, as we are still uncovering how life transitioned from entirely unicellular and microscopic to complex, multicellular life on Earth (see Chapter 5.4.3).

Discussion of the many factors affecting planetary habitability is expanded in the supplementary material.

7.2. Considerations for Habitable Environments

Thus far, the only known habitable environments are found on Earth. However, there are a number of potentially habitable environments scattered throughout the Solar System, many of which currently or in the past could resemble extreme environments on Earth (see Chapter 6.3). The rocky planets Mars and Venus are especially compelling targets for understanding habitability due to their similarity in size and location to Earth. There are several examples within the Solar System of uninhabitable planets with potentially habitable moons. The study of extrasolar planets and their parent stars also informs us about the formation and evolution of potentially habitable worlds. The similarity of planetary processes among diverse star systems suggests that moons orbiting exoplanets (*i.e.,* exomoons; Teachey and Kipping, 2018) are also worthy of consideration as potentially habitable environments, although no such bodies have been observed to date.

This section covers the state of our current understanding of the potential habitability of extraterrestrial environments in the Solar System and beyond. Several key bodies are the subjects of spacecraft missions in progress or in development: Mars (Chapter 7.2.1), Venus (Chapter 7.2.2), Europa (Chapter 7.2.3), Titan (Chapter 7.2.5), Ganymede (Chapter 7.2.6), and exoplanets (Chapter 7.2.10). These bodies are focal points of research activity and funding opportunities. We cover numerous other bodies because they have significance to the field of astrobiology even if they are not thought to be habitable.

7.2.1. Mars

Among Solar System bodies, Mars has received by far the most attention in the search for life beyond Earth. Exploration of Mars by multiple orbital and landed spacecraft has characterized global trends in the evolution of climate, aqueous activity, and geochemistry. Scientific consensus has tended toward less habitable surface conditions as the evolution of Mars progressed. In-depth exploration of Meridiani Planum, Gusev crater, and Gale crater by the rovers Opportunity (Arvidson *et al.,* 2014), Spirit (Ruff *et al.,* 2020), and Curiosity (Grotzinger *et al.,* 2014), respectively, has revealed regional and temporal variability in the potential habitability of the martian surface and near subsurface. Mars, like Earth, has hosted a wide diversity of environments that offer clues to understanding the habitability of Mars now or in the past.

7.2.1.1. Global trends in climate and aqueous activity. While volcanically generated basalt is the predominant type of rock on the martian surface, there is also evidence for the presence of liquid water throughout most of its history, with decreasing frequency through time (Kite, 2019). Because the presence of surface liquid water is critical for the habitability of a planetary surface, the climate has largely dictated the long-term surface habitability of Mars. The subsurface habitability of Mars is also dictated by climate, which impacts the crustal depths at which ice versus liquid water is stable, as well as the geochemical characteristics of any subsurface liquid water (Chapter 7.1.3 in the supplementary material). The climate of ancient Mars and the geochemical characteristics of surface and subsurface water can be determined by studying the geologic record of Mars.

The history of Mars is broken into three major eras (Fig. 7.2): Noachian (4.1–3.7 Ga), Hesperian (3.7–3.1 Ga), and Amazonian (3.1 Ga–present), each with different

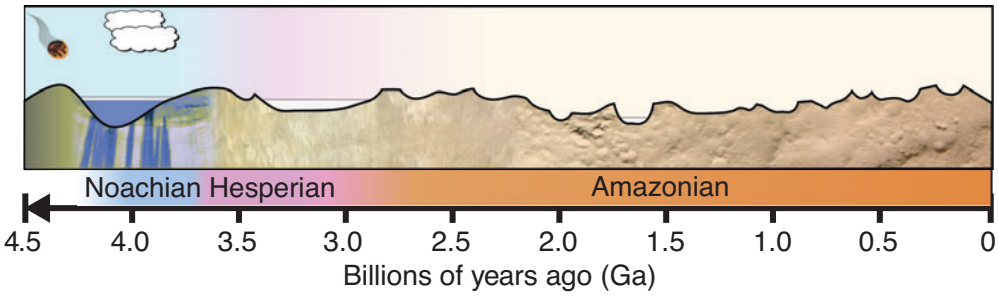


FIG. 7.2. Approximate timeline of the major martian geological eras. The Noachian (~4.1–3.7 Ga) is characterized by aqueous activity and an atmosphere more dense than today, though likely still colder than present-day Earth. The Hesperian (~3.7–3.1 Ga) is a transition period, showing signs of desiccation, atmospheric loss, and occasional aqueous activity. During the Amazonian (~3.1 Ga–present), Mars' surface has been cold, dry, and oxidized.

predominant conditions. During the Noachian, significant alteration of primary volcanic deposits formed clay and carbonate minerals on the surface and subsurface of Mars, which required the presence of liquid water (Bibring *et al.,* 2006). Transient warming episodes are thought to have occurred during the Hesperian, but with a lower frequency compared to the Noachian (Kite, 2019). Under such ''warm and wet'' periods, fluvial activity (*e.g.,* erosion by rivers) occurred during the Noachian and was particularly heightened during the Noachian–Hesperian transition (based on the relatively large number of fluvial channels visible in Noachian terrains from orbit; Fassett and Head, 2008). Even under warm and wet conditions, the climate of Mars was likely more similar to that in polar or alpine regions of Earth rather than terrestrial tropical climates at sea level. Groundwater and transient surface waters grew more saline during the Hesperian era, likely a reflection of conditions becoming even more cold and dry (Gendrin *et al.,* 2005; Ehlmann and Edwards, 2014). The Amazonian era (the current geologic era on Mars) is primarily characterized by the cold, dry conditions observed on the martian surface today, punctuated by heat and/or water delivery by impacts (Kite, 2019).

The climate of early Mars is still a subject of debate. To reconcile the geologic record of the martian surface with results from climate models, researchers have invoked seasonal melting (Palumbo *et al.,* 2018) and warming by transient, reducing greenhouse atmospheres (Wordsworth *et al.,* 2017; Turbet *et al.,* 2019). These scenarios imply that the surface of Mars was cold and icy for the majority of the Noachian era, but this cold was punctuated by greenhouse warming events that lasted 10^2 to 10^7 years (Kite, 2019).

Warming events could be triggered via release of CH_4 and H_2 formed by serpentinization reactions in the subsurface. This mechanism requires destabilization of methane clathrates in the martian cryosphere via one of three processes: (i) precession of axial tilt, since the obliquity of Mars is expected to have varied chaotically over timescales longer than 10^6 years (Touma and Wisdom, 1993; Kite *et al.,* 2020), (ii) large impacts (Chassefière *et al.,* 2016), or (iii) large-scale volcanic activity (Ramirez *et al.,* 2014; Chassefière *et al.,* 2016). While lakes and other potentially habitable surface environments may have existed early in Mars' history, during the Noachian era these climate-driven habitable environments would likely not be long-lived. Thus, any inhabiting life-forms would require refugia (see Chapter 7.1.5 in the supplementary material) to survive after transient, reducing greenhouse events had passed.

Study of the martian climate has suggested a dramatic loss of atmospheric gases over time, but the causes of this loss are not well understood (Jakosky, 2021). Loss of an atmosphere impacts habitability at the surface because the stability of liquids like water is reduced there. The main proposed mechanisms that may account for atmospheric loss on Mars are solar wind stripping due to lack of a protective magnetic field and impact erosion. These mechanisms are discussed in more detail in the supplementary material.

Some researchers have hypothesized that Mars may have had an ocean-sized body of water in its low-lying northern hemisphere (*e.g.,* Citron *et al.,* 2018), but this theory has substantial challenges. Most terrain north of 30°N on Mars is part of the northern lowlands, which is ~1–3 km lower in elevation than the more ancient southern highlands. Some have interpreted shoreline features to exist along the rim of this basin. Near this area, there is strong evidence for inland-sea-sized bodies of water on Mars (10^5 to 10^7 km^3 water, between the volumes of Earth's Caspian Sea and double the Mediterranean Sea), including the Eridania sea and seas within Valles Marineris. There is some evidence for submarine hydrothermal activity in the geologic record of Eridania basin, similar to subseafloor hydrothermal processes that supply energy for chemotrophic communities in Earth's oceans. Water drained into the northern lowlands of Mars from an inland sea in the Eridania basin in the late Noachian and from seas in Valles Marineris in the late Hesperian or early Amazonian. Water from both of these drainage events would have pooled in the northern lowlands, and some researchers have proposed that a long-lived northern ocean existed in this topographic low. However, geologic interpretations for the existence of this northern ocean are contested (*e.g.,* Salvatore and Christensen, 2014; Grant and Wilson, 2017), and the hypothesis of a long-lived northern ocean on Mars remains a matter of some debate.

7.2.1.2. Past transient habitable environments. A variety of martian environments have been studied for their potential to have been habitable in the past. Top among these environments are transient impact-generated hydrothermal systems, long-lived volcanically driven hydrothermal systems, and lakes that have been temporarily stable at the surface. Discussion of these environments is expanded in the supplementary material.

7.2.1.3. Proposed habitable environments for modern Mars. The current state of knowledge regarding possible habitable environments for extant life on Mars is summarized in Carrier *et al.* (2020) and references therein. Currently proposed habitable environments for extant life on modern Mars include within ice, within hypersaline brines or transiently hydrated salts, inside caves, and in the deep subsurface. Perchlorates (salts with a ClO_4^- anion) have been detected at $\sim$10 ppt concentrations by both the Phoenix lander and Curiosity rover. Perchlorate salts are strong oxidants (Lasne *et al.,* 2016), and saturated perchlorate brines have melting temperatures as low as $\sim$203 K (Hecht *et al.,* 2009). Microbes can reduce perchlorates for metabolism (Coates *et al.,* 1999). These microbes are found in arid environments on Earth, such as the Pilot Valley paleolake basin in Utah, indicating that any perchlorate-enriched brines on Mars could have sufficient redox energy to be habitable (Lynch *et al.,* 2019). The low melting temperatures and redox energy availability of perchlorate brines make them possible habitable environments on ancient and modern Mars, if these brines existed in the past as they do at present.

While habitable environments on the surface of Mars were likely transient and relatively short-lived, the subsurface of Mars may represent the longest-lived potentially habitable environment on the planet. A cold and icy Mars would have a cryospheric layer where all water, with the exception of some brines, would exist as ice. Beneath this insulating ice layer, geothermally heated liquid groundwater could persist, and chemolithotrophic microbial communities could potentially survive by harnessing redox energy produced through the same water–rock reactions that drive the deep subsurface biosphere on Earth (see Chapter 6.2.1.1). As the surface of Mars continued to get colder, the cryosphere would extend deeper, and potentially habitable subsurface environments would retreat to greater depths (Grimm *et al.,* 2017). Thus, the subsurface of Mars could have remained consistently habitable throughout its history and may continue to host microbial life today (Michalski *et al.,* 2013; Tarnas *et al.,* 2018).

Surface features known as recurring slope lineae (RSL; Fig. 7.3) observed by the Mars Reconnaissance Orbiter were initially thought to present potentially habitable environments on Mars (McEwen *et al.,* 2011). RSL are downward-moving darkened areas of hillsides coincident and recurrent with increased temperatures. However, the presence of liquid water in RSL is contested, and evidence suggests that RSL can be explained without the presence of liquid water (Dundas *et al.,* 2017; Schmidt *et al.,* 2017). RSL remain under ''special region'' status in NASA's planetary protection policies due to their potential habitability (see Chapter 10.3.1).

Ultimately, Earth-like organisms would require drastic adaptation to survive in surface environments on Mars, or in near-surface ice, brine, hydrated salt, or cave environments connected to Mars' desiccating and low-pressure atmosphere (Carrier *et al.,* 2020). The deep subsurface of Mars has potential for hosting Earth-like organisms, but only where groundwater is present. Because the presence of groundwater on Mars has not been unambiguously demonstrated, it is unclear whether a deep subsurface habitable environment exists on Mars today. See Chapter 9.3.1.2 for considerations regarding life as we don't know it on Mars.

7.2.2. Venus

Modern Venus is a hostile place for life as we know it. Despite their proximal orbits and similar planetary origins, Venus and Earth have followed different evolutionary pathways. The surface of Venus is not considered to be habitable today, although it may have been habitable in the past. Understanding of venusian history is currently limited by the extremely hostile environments on the surface and within the atmosphere that greatly hamper measurements of composition and geological characteristics. However, Venus serves as a vital laboratory for exploring planetary evolution processes. A radar map showing the roughness of Venus's surface, which correlates with geological provinces, is shown in Fig. 7.4. Comparing Venus to other planets helps constrain climate models, refine concepts such as the stellar habitable zone (Chapter 7.1.4.1 in the supplementary material), and identify possible futures for Earth.

7.2.2.1. Atmospheric structure. Venus experienced a runaway greenhouse state (see Chapter 3.4.1.4). The dense,

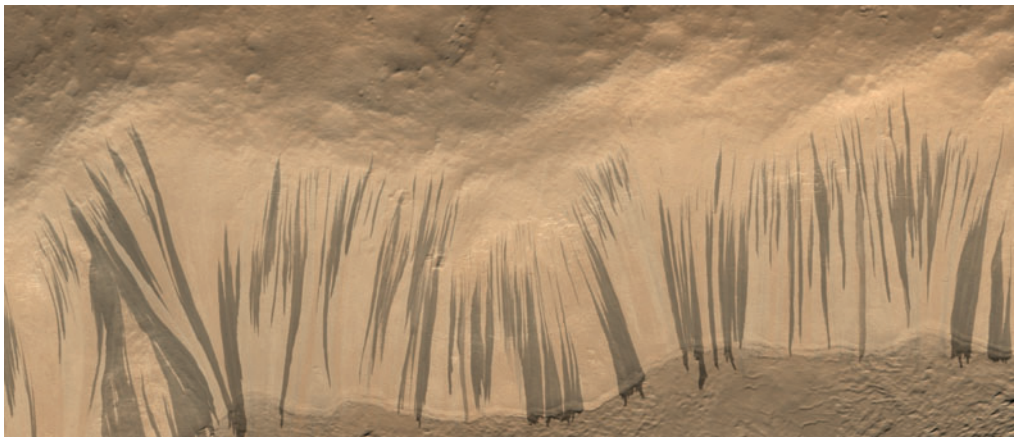


FIG. 7.3. Recurring slope lineae (RSL) on the surface of Mars. North is roughly to the left in this image, and the Sun is shining from the bottom. More details regarding this HiRISE image are available at https://www.uahirise.org/PSP_001656_2175. Credit: NASA/JPL/University of Arizona.

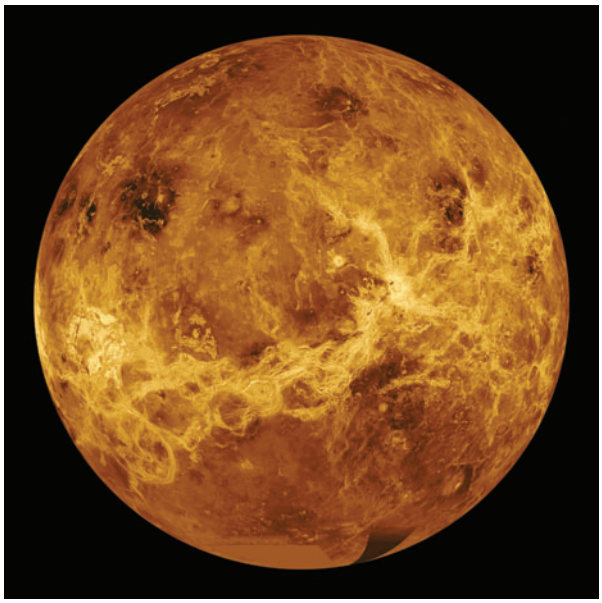


FIG. 7.4. False-color image of the surface of Venus, as seen by radar from the Magellan and Pioneer Venus Orbiter missions. A wide array of geologic features are identifiable through differences in surface roughness—more rough materials appear bright from scattering a greater fraction of radio waves back to the receiver than do smooth materials. Credit: NASA/JPL-Caltech.

CO_2-rich atmosphere and H_2SO_4 in the clouds are now opaque to nearly all thermal wavelengths for temperatures relevant to the atmosphere of Venus, meaning thermal radiation emitted from the surface and lower layers of the atmosphere is constantly absorbed and re-emitted in all directions. The interior of Venus is heated by radioactive decay (similar to Earth), and this heat can therefore escape only very slowly through absorption/re-emission of thermal radiation and by atmospheric convection. In contrast, other rocky bodies cool by emitting thermal radiation directly to space. Due to the high altitude and opacity of the cloud deck on Venus, radiative cooling is limited to the top of the atmosphere (Haus *et al.,* 2015). These effects form a steep adiabatic temperature gradient between the cloud tops and the surface, ultimately resulting in an average surface temperature of about 464°C (737 K).

At present, there is very little water in the venusian atmosphere—10 ppmv or less above the cloud deck (Marcq *et al.,* 2018) and about 30 ppmv below (Arney *et al.,* 2014). Lack of geologic exploration severely limits our ability to establish a timeline for the evolution of Venus (Taylor *et al.,* 2018). The ratio between deuterium, ^{2}H or D, and hydrogen, ^{1}H, typically called the D/H ratio, is a useful probe of a planetary body's history due to its connection to various geophysical processes that can change the planetary water budget. The D/H ratio in the atmosphere of Venus is considerably higher than that present in seawater on Earth, with estimates ranging from 95 to 240 times that of Earth (Krasnopolsky *et al.,* 2013), possibly increasing with altitude throughout the cloud deck (Fedorova *et al.,* 2008). Because of their presumed proximity in the solar nebula during planetary formation, Venus and Earth are thought to have had similar primordial compositions, including D/H ratio (Morbidelli *et al.,* 2012). Venus is therefore thought to have also had a primordial water inventory somewhat comparable to that of Earth, up to around a 500 m global average depth (Donahue and Russell, 1997).

The high D/H ratio on Venus is thought to have been the result of enrichment processes, primarily attributed to thermal escape (sometimes called ''Jeans escape'') following photolysis of H_2O vapor in the upper atmosphere (Krasnopolsky *et al.,* 2013). Venus is believed to have entered a ''moist greenhouse,'' where a hot atmosphere with a high volume fraction of H_2O vapor causes this vapor to reach high enough altitudes to be photolyzed by solar EUV photons (Kasting, 1988). H_2O molecules, which occasionally have D in place of H, are destroyed by absorbing the high-energy EUV photon. This process releases D and H, which recombine to form DH and H_2 molecules. As on Earth, the gravity of Venus is weak enough to permit H_2 to escape at ambient temperatures at the exobase (the outermost boundary of the viscous atmosphere; see Chapter 3.4.1.5). However, the heavier DH molecules are lost more slowly than H_2 due to their higher mass, resulting in an enrichment of D/H.

Similar D/H enrichment processes happen much more slowly on Earth because water vapor condenses in the lower troposphere, typically well below the UV-absorbing ozone layer. Cloud condensation on Earth therefore acts as a ''cold trap'' (Holton and Gettelman, 2001), keeping the water close to the surface. Conditions at Venus could have resulted in the loss of an Earth-ocean-equivalent amount of water (Kasting and Pollack, 1983). Estimates vary on just how much water Venus has had, and lost (Wordsworth and Pierrehumbert, 2013). Because a moist greenhouse state can lead to rapid water loss, its onset is often defined to be the ''pessimistic'' inner edge of the habitable zone (Wolf and Toon, 2015). A moist greenhouse state is typically expected to lead to a runaway greenhouse following the loss of water.

The D/H ratio in the atmosphere can also be altered by other means, complicating the interpretation of the high D/H ratio of Venus (Grinspoon, 1993). Hydration of minerals by chemical weathering removes water from the atmosphere of a body, potentially storing water within the crust that may be reintroduced via volcanic outgassing. Chemical weathering reactions may also proceed more readily with a lighter or heavier isotope, contributing a secondary fractionation effect.

Discussion of Venus's atmospheric structure and lack of a magnetic field is expanded in the supplementary material.

7.2.2.2. Possible histories for Venus. Although modern Venus is very desiccated, its elevated D/H ratio has important implications for possible past habitability. Venus has lost the vast majority of the water it once contained, but the lack of a geologic record that may provide evidence for how long this water loss occurred makes it impossible to say whether the surface or atmosphere of Venus was ever habitable (Taylor *et al.,* 2018). Venus may never have had liquid water on its surface. Such a scenario would make venusian origins of life more difficult, although a cloud-based origin of life is conceivable (*e.g.,* Dobson *et al.,* 2000). If the atmosphere of Venus never cooled enough to

condense water on its surface, it would likely have lost much of its primordial water rapidly, within 100 million years (Hamano *et al.,* 2013).

If Venus cooled enough to condense liquid water on its surface, it is possible that life could have emerged under conditions similar to origins on Earth. Life exhibits an ability to adapt and expand to fill even extreme habitats (see Chapter 6.3). Furthermore, a ''faint young Sun'' (Chapter 4.1.4) likely resulted in lower equilibrium temperatures on the surface of early Venus. General circulation models incorporating atmospheric feedback show that if Venus has been rotating slowly (taking longer than 16 Earth days per revolution) for most of its history, its surface may have remained hospitable until as recently as 0.715 Ga (Way *et al.,* 2016). Venus may have experienced chemical weathering and subsequent carbon burial that provided a negative climate feedback as the young Sun warmed, helping to maintain a habitable environment on the surface of Venus over geologic timescales (Zolotov, 2019). If the surface environment at Venus gradually warmed to evaporate an Earth-sized ocean of water, organisms in that ocean may have had time to adapt to changing conditions and inhabit the clouds (*e.g.,* Limaye *et al.,* 2018), although such a scenario faces significant challenges.

Greaves *et al.* (2021) identified an absorption feature in observations of Venus that they attributed to phosphine (PH_3) in the cloud deck, which they claimed as a potential biosignature. However, this interpretation is contested. See Chapter 8.3.1.3 for more on the possibility of phosphine on Venus.

The crust of Venus exhibits widespread consistency in its surface age, ranging from 0.3 to 0.7 Ga, as inferred from crater densities (Solomon *et al.,* 1999). The broad consistency of surface age implies that a major catastrophe triggered a global resurfacing event (Strom *et al.,* 1994). Possible explanations for such an event are described in the supplementary material.

At present, it is difficult to reconcile any scenario for the evolution of Venus as a planet that may still be habitable for even the most extreme and adaptive of Earth-like life. Still, Venus presents an excellent laboratory for better understanding atmospheric evolution and surface–atmosphere interactions under more extreme conditions than those on Earth. If Venus was ever habitable, similar processes could be operating on exoplanets under similar conditions, and studying Earth's nearest neighbor can provide insights into the habitability of such exoplanets. See Chapter 9.3.1.3 for considerations regarding life as we don't know it on Venus.

7.2.3. Europa

Jupiter's moon Europa (Fig. 7.5) is an especially attractive target in the study of habitability. Evidence supports the presence of an ocean over 100 km deep with an overlying ice shell over 20 km thick. The large orbital eccentricity of Europa, maintained by orbital resonances with Io and Ganymede, causes extensive tidal heating that helps maintain a liquid ocean (Tobie *et al.,* 2003). Silicate rock in contact with the europan ocean is likely to provide a source of chemical energy for life. Chemical reactions and leaching interactions between Europa's ocean and mantle may have altered the composition and oxidation state of the ocean, which have important consequences for habitability and possible origins of life.

7.2.3.1. Evidence for an ocean. The strongest evidence for a subsurface europan ocean comes from magnetic sounding (Khurana *et al.,* 2009), with additional support provided by geological features (*e.g.,* Figueredo and Greeley, 2004). Jupiter applies an oscillating magnetic field to Europa. An induced magnetic field that oscillates along with Jupiter's rotation has been observed from Galileo magnetometer measurements near Europa (Kivelson *et al.,* 2000), and the best explanation of these measurements is a liquid water ocean with a salinity roughly comparable to oceans on Earth (Hand and Chyba, 2007). Other materials cannot explain the induced field observed near Europa. A more detailed discussion is found in the supplementary material.

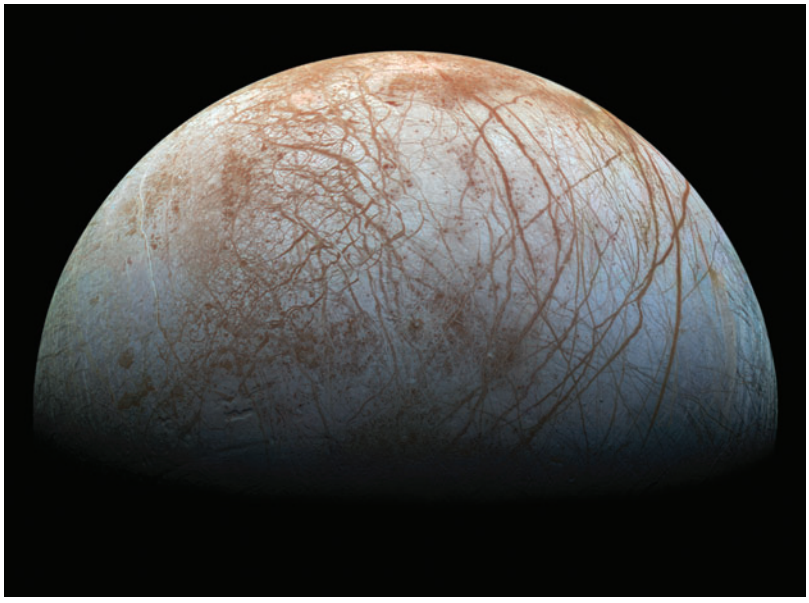


FIG. 7.5. The surface of Europa, as imaged by the Galileo spacecraft. Cycloid ridges and chaos terrains imply recent geologic activity that helps us understand the potential habitability of this intriguing world. Credit: NASA/JPL-Caltech/SETI Institute.

7.2.3.2. Surface conditions. Europa rests deep within the extensive magnetosphere of Jupiter. The strong jovian magnetic field traps charged particles from Io and Jupiter (Bagenal *et al.,* 2015). These trapped particles corotate with Jupiter, causing them to overtake Europa with a relative speed of 100 km/s. Europa has a negligible atmosphere (~0.01 nbar; Kliore *et al.,* 1997), so these particles impact the surface, sputtering ice and causing radiolytic chemical reactions. Plasma impacts the entire surface of Europa, but not in a spatially uniform way. The high-radiation environment at the surface of Europa renders it unfit for life, but this sterilizing effect likely only penetrates 1–20 cm depending on latitude and longitude (Nordheim *et al.,* 2018).

Estimates for the thickness of the europan ice shell range from a minimum of about 8 km needed to support the observed impact craters (Schenk and Turtle, 2009) to as much as 35 km inferred from thermodynamic models (*e.g.,* Vilella *et al.,* 2020). A wide variety of studies converge on the range 19–25 km (*e.g.,* Howell, 2021) with a cold, brittle, thermally conductive ice layer overlaying a warmer, ductile, thermally convective ice layer (Pappalardo *et al.,* 1998). The surface of Europa (Fig. 7.5) is covered in tectonic features reminiscent of subduction faults and fractures common on Earth (Leonard *et al.,* 2018). There is some evidence of subduction (Kattenhorn and Prockter, 2014), although features may be created by interaction between the conductive and convective ice layers, mediated by tidal forces (Barr and Hammond, 2015).

Some evidence suggests that plumes may be erupting from the surface of Europa. Observations with the Hubble Space Telescope (Roth *et al.,* 2014b) and *in situ* measurements from Galileo (Jia *et al.,* 2018; Arnold *et al.,* 2019) each show some indications of water vapor lofted to high altitudes. Later observations with Hubble have seen no further evidence of plumes, suggesting that europan plumes can only be intermittent, if they are indeed present (Roth *et al.,* 2014a). Further studies have had difficulty corroborating these observations—there does not appear to be any thermal anomalies on the surface that would be expected where the plumes erupt (Rathbun and Spencer, 2020), a stark contrast with the plumes of Enceladus (Porco *et al.,* 2014). If Europa does have plumes, they are likely to contain material not from the ocean but from shallower melt or brine pockets within the ice shell, as fractures are unlikely to reach directly from the ocean to the surface (Rudolph and Manga, 2009; Howell and Pappalardo, 2019).

7.2.3.3. Ocean habitability. Europa seems to have many of the conditions required to support habitable environments. Hydrothermal vents likely exist in the subsurface ocean of Europa, providing a possible energy source for metabolism (Lowell and DuBose, 2005; Vance and Melwani Daswani, 2020). If there is a transport mechanism through the ice shell (*e.g.,* Kalousová *et al.,* 2014), introduction of oxidized surface materials may provide rich sources of chemical energy for the likely reduced ocean (Fig. 7.6; Vance *et al.,* 2019; Hand *et al.,* 2020). Pockets of concentrated brine may also become trapped in the ice shell, thereby providing environments suitable for psychrophilic (Chapter 6.3.1.1) and halophilic (Chapter 6.3.3.1) organisms

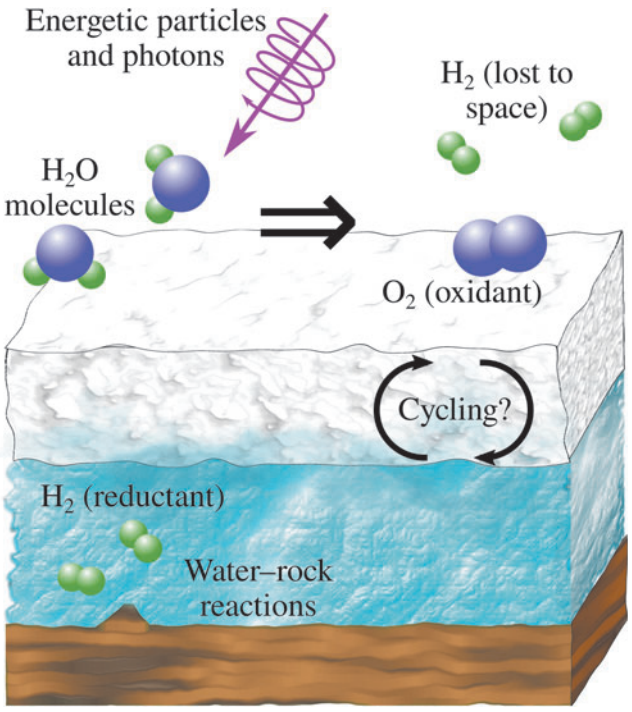


FIG. 7.6. Sources of biologically important molecules that could potentially sustain life in Europa's ocean. The high-radiation environment at the surface acts as a source of oxidants, and water–rock reactions at the seafloor may act as a source of reductants. If there is some process that cycles crustal material between the surface and the ice–ocean interface, similar to plate tectonics on Earth, this interface likely represents a habitable environment. Europa's ice shell is expected to be quite thick (over 20 km), which may preclude such cycling.

(Boetius *et al.,* 2015; Cooper *et al.,* 2019). Considerations for life as we don't know it on Europa are discussed in Chapter 9.3.2.2.

Subduction on Europa likely cannot cycle surface material directly into the ocean (Howell and Pappalardo, 2019). This has important consequences for studies of the habitability of the europan ocean that rely on transport of oxidants from the surface into the ocean (*e.g.,* Hand *et al.,* 2009; Vance *et al.,* 2016). There is mounting evidence that the europan ice shell is in the process of thickening and cooling (Leonard *et al.,* 2018; Allu Peddinti and McNamara, 2019; Howell and Pappalardo, 2019). The thickness and properties of the ice shell also dictate how energy from tidal interactions will be distributed (Matsuyama *et al.,* 2018), so a thorough understanding of the dynamics of the europan ice shell is critical to understanding the habitability of the ocean.

The location of Europa in the Solar System and its bulk density suggest that it is likely constructed of rock, ice, and metals sourced from CI chondrite material (Sohl *et al.,* 2002), although there may be an additional cometary component (Néri *et al.,* 2020). Laboratory analog research on leaching of salts from chondritic material suggests $MgSO_4$ may be a primary salt in the europan ocean (Kargel *et al.,* 2000). This is supported by putative detection of $MgSO_4$ spectral features on the surface (McCord *et al.,* 1998; Fox-Powell *et al.,* 2019), although other materials may explain the observed spectra (Hibbitts *et al.,* 2019), such as NaCl

that may come from the subsurface (Trumbo *et al.,* 2019). Some studies (*e.g.,* Brown and Hand, 2013) suggest that surface S and O may be implanted exogenously from volcanic activity on Io, and their presence on the surface does not preclude an ocean dominated by chlorides such as NaCl.

7.2.3.4. Upcoming exploration. An overview of the upcoming Europa Clipper and JUICE missions is included in the supplementary material.

7.2.4. Enceladus

Enceladus is a small (504 km diameter), icy moon of Saturn. Despite its small size, Enceladus has received abundant scientific attention, including measurements from the NASA Voyager I, Voyager II, and Cassini missions. This moon is of particular astrobiological interest, as there is strong evidence of a global ocean below its icy surface that is in contact with a rocky core.

The existence of a subsurface ocean on Enceladus is supported by several lines of evidence. Images taken by Cassini show a physical libration (wobble) of the icy crust that is too large for an icy shell anchored to the silicate core of the moon but is consistent with a subsurface liquid ocean (Thomas *et al.,* 2016). Gravity and shape data of the moon indicate the icy crust may be tens of kilometers thick on average but less than 11 km thick at the southern pole, with a liquid ocean beneath (Beuthe *et al.,* 2016; Čadek *et al.,* 2016). Numerous persistent jets of water vapor and small icy particles emanating from the ''tiger stripes'' region around the southern pole further support the presence of a subsurface ocean (Fig. 7.7; *e.g.,* Porco *et al.,* 2006; Waite *et al.,* 2006; Postberg *et al.,* 2009, 2011).

The composition of these plumes was sampled directly by the Cassini spacecraft with multiple plume fly-throughs, in addition to *in situ* samples and measurements of the E-ring of Saturn, which is sourced from the plumes of Enceladus (Spahn *et al.,* 2006). A variety of grains were observed in the plumes, with compositions ranging from salt-rich (Postberg *et al.,* 2009; Khawaja *et al.,* 2019) to water-dominated (Waite *et al.,* 2006; Teolis *et al.,* 2010), to grains rich in organic molecules or silicates (Postberg *et al.,* 2008, 2018). In addition to the grain composition, Cassini instruments measured the plume gas composition and found it to be predominantly H_2O, along with significant CO_2, CH_4, NH_3, and H_2 (Waite *et al.,* 2006, 2017). These measurements indicate the icy plumes are likely sourced from a subsurface liquid ocean in contact with a silicate core. Alternative explanations for the source of the plumes, discussed in the supplementary material, have been considered, but none are consistent with observations.

To remain consistent with observed salt and NH_3 measurements, the temperature of the subsurface ocean may be above −1°C on average (Glein *et al.,* 2018). Similarly, from constraints of CO_2 in the plume, NaCl and bicarbonate particle abundances, and the presence of silica nanograins in the E ring of Saturn, the pH of the plume source is expected to be between 9 and 11 (Cable *et al.,* 2020). The source of the plumes may be derived from high-temperature hydrothermal activity at the water–rock interface, allowing the average pH of the ocean to be lower, perhaps between 8 and 9 (Sekine *et al.,* 2015; Glein and Waite, 2020). Thus, both the temperature and pH of the ocean may be suitable for life (see Chapter 7.1.1.1 in the supplementary material). A source of chemical energy may be available in the form of H_2 and CH_4 generated from serpentinization of the silicate core and radiolysis of water in the seafloor pore space (Bouquet *et al.,* 2017). The combination of all these factors indicates that the enceladean ocean may be habitable to life as we know it.

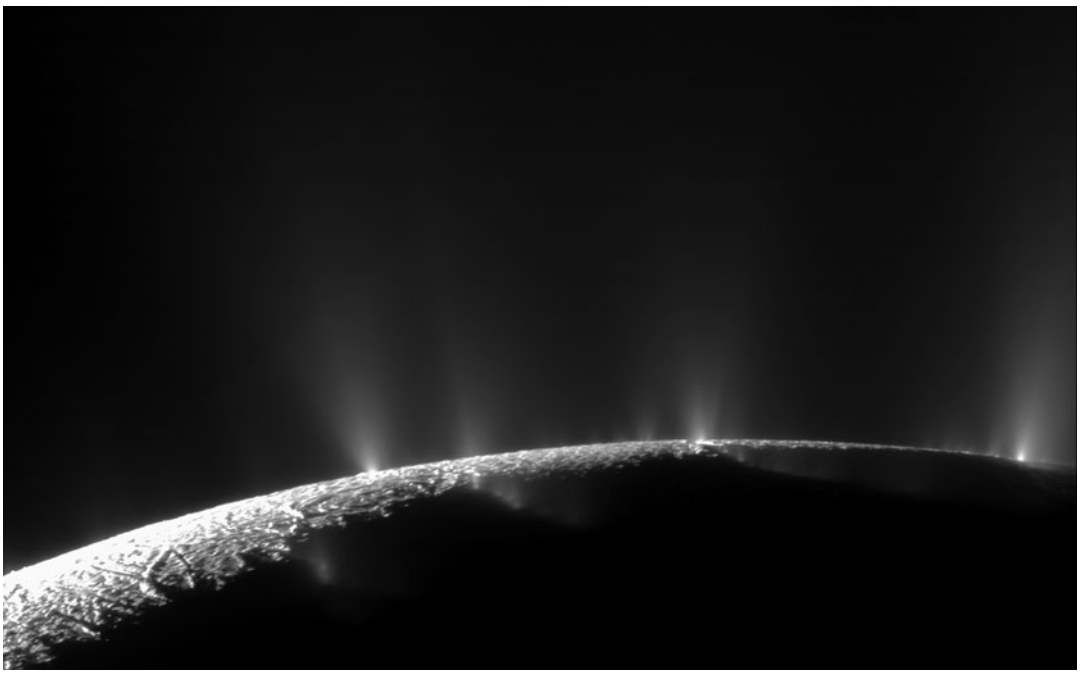


FIG. 7.7. The south polar plumes of Enceladus, lit by the Sun from behind the camera of the Cassini spacecraft. Ice grains from subsurface reservoirs of liquid water are ejected into space and reflect the sunlight. Credit: NASA/JPL-Caltech/SSI.

It is unknown if the conditions for the origin of life, either at present or in the past, have been met on Enceladus. Even if the conditions for life and abiogenesis are present in the enceladean ocean, the moon may be too young to have formed a biosphere. Models and measurements of saturnian moons and rings indicate they may be only a few hundred million years old or less (Ćuk *et al.,* 2016; Iess *et al.,* 2019), although this proposal is debated (Crida *et al.,* 2019; Neveu and Rhoden, 2019). If Enceladus is old, its porous interior may have already been completely serpentinized, halting the production of chemical energy sources (Zandanel *et al.,* 2021). Future missions to this icy moon are necessary to confidently detect life in the subsurface ocean, should it exist. For a more in-depth consideration of Enceladus and its astrobiological relevance, see the recent review by Cable *et al.* (2020). Considerations for the possibility of life as we don't know it on Enceladus are discussed in Chapter 9.3.2.3.

7.2.4.1. Comparison between Enceladus and Europa. Enceladus and Europa share a number of key similarities and differences worthy of note. Both are icy moons known to harbor subsurface liquid water oceans that may be abodes for life. Both Europa (Anderson *et al.,* 1998) and Enceladus (Neumann and Kruse, 2019) are likely differentiated. Europa's bulk density and axial moment of inertia imply that it is mostly silicate rock, with a possible iron core and a water + ice layer with a thickness close to just 10% of the body radius (Chapter 7.2.3.2). Enceladus, in contrast, likely has a porous core of silicates and a water + ice layer with a thickness of up to about 25% of the body radius. Cassini sampling of the plumes of Enceladus implies a high ocean pH as described above. A variety of ocean compositions and pH values have been postulated for Europa's ocean (Zolotov and Kargel, 2009), but there is insufficient evidence to constrain it as yet. Europa's surface is more heavily irradiated than that of Enceladus, thus causing increased oxidation of surface materials from O liberated by photolysis of H_2O (Parkinson *et al.,* 2008; Hand *et al.,* 2009).

Like that of Enceladus, Europa's surface is mostly water ice (Carlson *et al.,* 2009), but Europa has a large amount of darker material on its surface as well, likely sourced from volcanoes on Io or other endogenous sources (Hand and Carlson, 2015; Hibbitts *et al.,* 2019). In contrast, Enceladus has a very bright surface consistent with pure ice plus a small amount of NH_3 and tholins in a global coating spread by the action of plumes (Hendrix *et al.,* 2010). Many lines of evidence have confirmed the presence of plumes at Enceladus, including direct images (Fig. 7.7), but only intermittent plume activity is expected on Europa as described above. Alternative explanations may account for existing observations instead of plumes (Roth, 2021; Styczinski and Harnett, 2021). Europa is much larger than Enceladus (radii approximately 1560 km and 252 km, respectively), with a higher surface gravity. The ice shell of Enceladus may be as thin as a few kilometers at the south pole (Hemingway and Mittal, 2019), while Europa's stronger gravity likely prevents any conduit from reaching all the way from the ocean to the surface (Rudolph and Manga, 2009).

As Europa and Enceladus are both thought to have liquid water oceans in contact with silicate rock, they are both promising candidates in the search for life elsewhere. Each may be habitable. What we learn about one moon can sometimes help us understand the other, but we must always keep in mind that Europa is not Enceladus, and vice versa, and exercise caution in studying these bodies.

7.2.5. Titan

Titan is the largest moon of Saturn, with a radius of 2575 km (50% larger than Earth's moon), and it is the only moon in the Solar System with a significant atmosphere. Due to its thick atmosphere (1.5 bar surface pressure) and accompanying haze particles, the surface of Titan remained obscured until the early 2000s. The arrival of the joint NASA/ESA Cassini–Huygens mission to Saturn in 2004 and the successful landing of the Huygens probe on the surface of Titan in 2005 provided unprecedented details about its surface, atmosphere, and photochemistry. That mission revealed that Titan is the only body in the Solar System other than Earth with standing bodies of liquid at the surface (Hörst, 2017), bodies that are likely rich in methane or ethane (Brown *et al.,* 2008). Figure 7.8 shows an image of Titan in front of Saturn, taken by Cassini.

Titan's atmosphere is mostly N_2, with up to about 5% CH_4 at the surface (Hörst, 2017). Measurements from the Cassini–Huygens mission found that large ions, potentially organic molecules, were present in the upper atmosphere (*e.g.,* Coates *et al.,* 2007; Crary *et al.,* 2009). The exact makeup of these large molecules will require future measurements. Photochemical models of the titanian atmosphere and surface chemistry imply that complex organic molecules such as nucleotide bases and amino acids may be formed high in Titan's atmosphere (Fig. 7.9; Hörst *et al.,* 2012). These organic molecules would fall to the surface, forming the observed dunes, lakes, and seas. The NASA

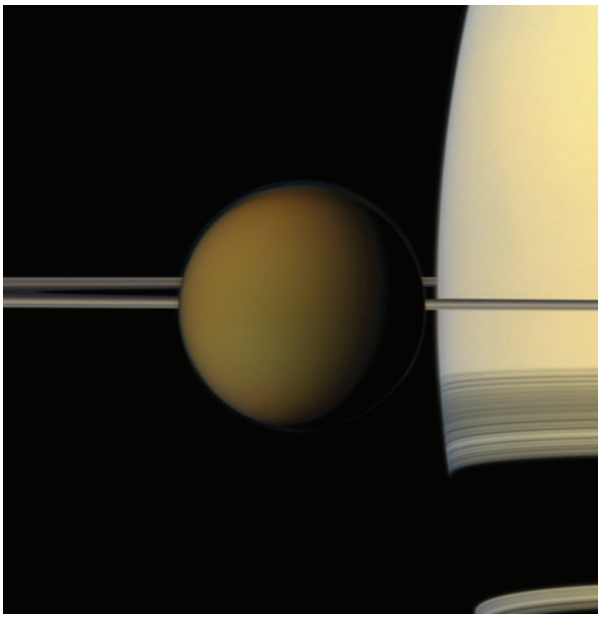


FIG. 7.8. Titan in front of its parent planet Saturn, a natural-color image taken by the Cassini spacecraft. The hazy appearance is the result of the organic-rich haze particles in the atmosphere. Credit: NASA/JPL-Caltech/Space Science Institute.

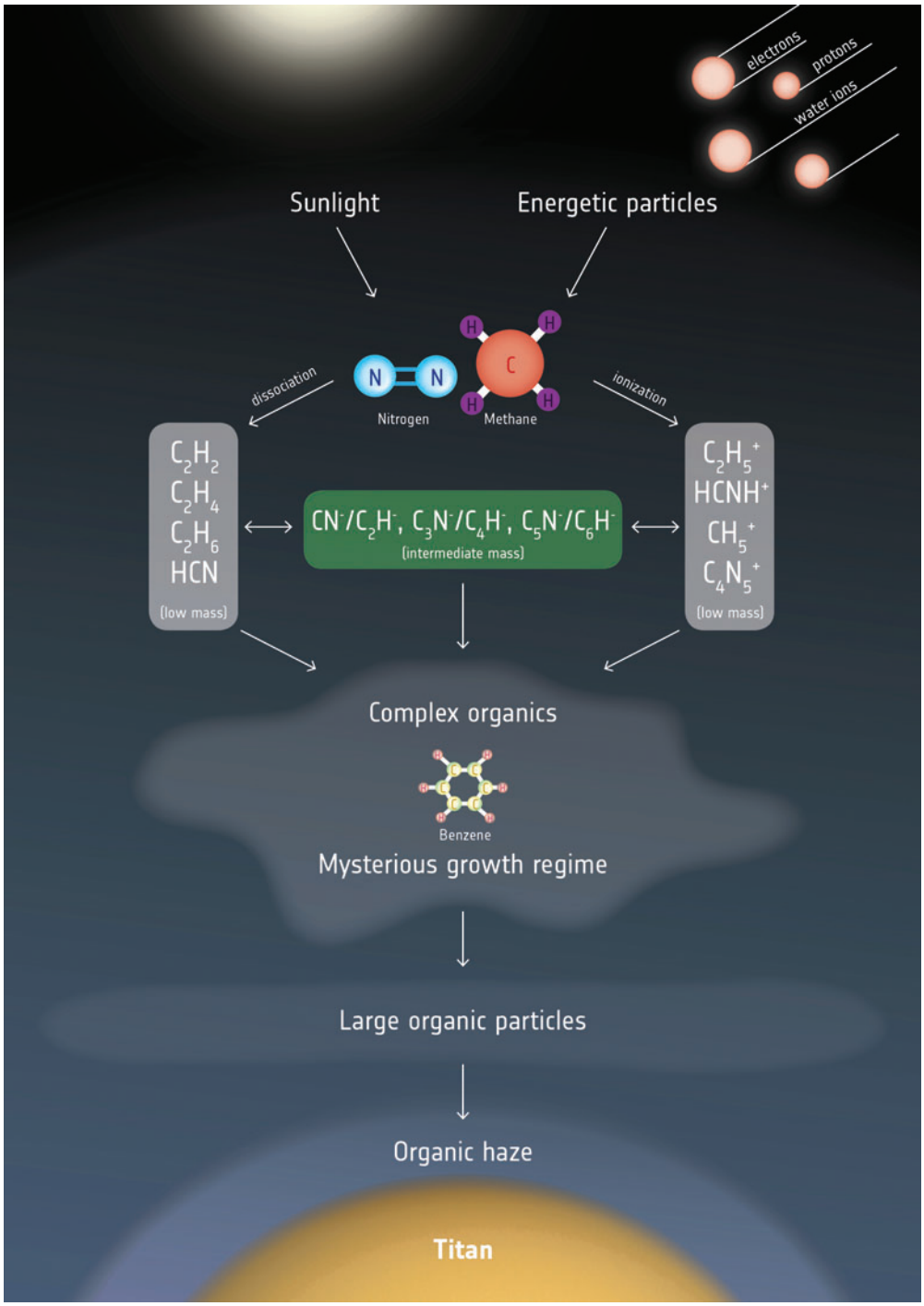


FIG. 7.9. Photochemistry in Titan's atmosphere. Energetic photons and charged particles interact with methane molecules that diffuse to high altitudes, freeing hydrogen. The resulting ions recombine to form a wide variety of complex molecules, possibly including precursors to life like amino acids. The large molecules in Titan's atmosphere create hazes that obscure the surface. Credit: ESA.

Dragonfly mission, scheduled to launch in June 2027, will explore the surface of Titan to characterize the environment and search for signs of life.

While it is uncertain if life could exist in Titan's surface lakes and seas, its subsurface environments may provide a habitable environment suitable for Earth-like life, similar to other icy moons in the Solar System (Lunine *et al.,* 2020). Measurements from the Cassini–Huygens mission indicate that a subsurface ocean is present beneath the icy shell of Titan (Béghin *et al.,* 2010; Iess *et al.,* 2012; Mitri *et al.,* 2014), although little is known about its composition. Cryovolcanism or impact-induced melting could also lead to transient habitable conditions in surface ices, such as within melt pockets (O'Brien *et al.,* 2005). Those conditions might have provided sufficient time for biologically relevant chemistry to arise (Neish *et al.,* 2008, 2010) but might have

provided insufficient time for the formation of life (see Chapter 7.1.5.4 in the supplementary material). See the review by Lunine *et al.* (2020) for further discussion of the astrobiological relevance of Titan. See Chapter 9.3.2.1 for a discussion of such life as we don't know it on Titan.

7.2.6. Ganymede and Callisto

Jupiter's two largest moons, Ganymede and Callisto, share many characteristics and are both of astrobiological interest. Ganymede is widely believed to possess a subsurface ocean. Callisto may also possess an ocean. Each moon is larger than Earth's moon and has a low density, consistent with large quantities of water + ice in their makeup (Schubert *et al.,* 2004). Unlike Ganymede, Callisto is likely not fully differentiated (Anderson *et al.,* 2001), but dominant ice at the surface suggests that it has differentiated enough to form an ocean that may persist today. Three characteristics common to both moons are influential to considerations of their possible habitability: their ancient surfaces, presence of magnetic induction signals, and high-pressure ices deep in their interiors.

The surfaces of Ganymede and Callisto are extensively cratered, to the point of saturation in many areas (Pappalardo *et al.,* 2004). A wide variety of organic molecules have been identified on the surfaces of both moons (McCord *et al.,* 1997). These molecules may have been emplaced by exogenous sources, such as impacts, or endogenous sources, such as tectonic activity. However, both moons have very old surfaces. Callisto exhibits no apparent endogenic activity, suggesting a surface age over 4 billion years everywhere. Ganymede's surface age is probably at least 1 billion years everywhere, with some regions as much as 4 billion years old (Showman and Malhotra, 1999). Although the younger terrain on Ganymede may have been resurfaced at some point in that moon's history, it must have taken place very long ago (Showman *et al.,* 2004). Therefore, cycling of material between the surfaces and possible oceans of these bodies likely does not happen, and surface composition may not be informative of ocean composition.

Ganymede has an intrinsic magnetic field that creates a mini-magnetosphere, largely shielding its surface from Jupiter's intense radiation belts (Kivelson *et al.,* 1997). Galileo measurements (Kivelson *et al.,* 2002) and UV images of Ganymede's auroras (Saur *et al.,* 2015) are consistent with the presence of an ocean, but definitive proof of a subsurface ocean is lacking. Callisto shows no intrinsic magnetic field, but instead it shows a relatively strong induced magnetic field (Zimmer *et al.,* 2000). Initially, this condition was attributed to a subsurface ocean, but research examining the plasma environment has suggested that induction purely in Callisto's ionosphere may be able to account for the entire observed field (Hartkorn and Saur, 2017; Liuzzo *et al.,* 2017).

Water–rock interactions are likely critical to the possible habitability of Ganymede and Callisto. However, these moons are so large and the pressures at depths consistent with their likely oceans are so great that exotic phases of water are formed, known as high-pressure ices (Journaux *et al.,* 2020). Layers of ice below the oceans could be detrimental to both origin-of-life scenarios and ongoing habitability, as they may stifle chemical interaction at the water–rock interface. However, the properties of high-pressure ices, especially for relevant ocean compositions, are not understood well enough to rule out complex dynamics such as upward ''snow'' in the presence of high salinity (Vance *et al.,* 2014). Convection in the high-pressure ices may also facilitate contact between the ocean and silicates (Kalousová *et al.,* 2018). Heat escaping from the interior may drive the formation of dense brine layers that underlie the high-pressure ices and sustain water–rock contact (Vance *et al.,* 2018), potentially providing abodes for life.

Although the structures of both Ganymede and Callisto present challenges to conceiving of them as habitable worlds, each may yet contain a subsurface ocean with the ingredients needed to sustain life. If the interiors of these icy ocean worlds support life, life may be quite common indeed.

7.2.7. Other icy worlds

7.2.7.1. Ceres. Ceres is the only extant dwarf planet in the asteroid belt, making it the closest dwarf planet to the Sun. It has astrobiological significance due to the potential presence of past and/or presently habitable conditions (Castillo-Rogez *et al.,* 2020). Results from the Dawn mission to Ceres (2015 – 2018) demonstrated that this water-rich body has experienced a dynamic history of water–rock interactions, which likely are still occurring today. The spectral identification of phyllosilicates and carbonates on Ceres demonstrates that primary volcanic minerals were aqueously altered (Milliken and Rivkin, 2009; De Sanctis *et al.,* 2016; Carrozzo *et al.,* 2018). Small deposits of organic molecules have also been identified on the surface of Ceres (De Sanctis *et al.,* 2017).

There is evidence that Ceres may currently have liquid water in the subsurface. Occator crater on Ceres contains evidence for recent—perhaps ongoing—surface brine emplacement, spurred by its parent impact event (De Sanctis *et al.,* 2020; Raymond *et al.,* 2020). Evidence for the presence of a subsurface brine pocket beneath Occator crater comes from gravity anomaly mapping (Raymond *et al.,* 2020), spectral identification of hydrated sodium chloride on the Cerealia Facula dome (De Sanctis *et al.,* 2020), and surface morphological expressions of recent aqueous activity (Nathues *et al.,* 2020; Schenk *et al.,* 2020; Schmidt *et al.,* 2020; Scully *et al.,* 2020), all acquired during Dawn Extended Mission 2. Brine reservoirs can be preserved on this relatively small, heat-deprived body if $\sim$35% or more of the crust is composed of methane clathrate hydrate, which is a strongly thermally insulating material (Castillo-Rogez *et al.,* 2019). Prospects for a habitable environment in the subsurface of Ceres are certainly higher than in any other object in the asteroid belt, and clear evidence for past water–rock interactions also indicates that Ceres could have previously been habitable.

7.2.7.2. The moons of Uranus. Uranus has five large moons. From nearest to farthest from the planet, they are Miranda, Ariel, Umbriel, Titania, and Oberon, all of which orbit near the rotational equator of Uranus. They are likely natural satellites, formed from the same planetary nebula as the planet. Uranus orbits the Sun at a distance of nearly 20 AU, so this frigid system receives nearly 400 times less

insolation than Earth. However, evidence for an ocean in Pluto's subsurface (Chapter 7.2.7.4) suggests that the habitability of one or more uranian moons is plausible. All we know of the Uranus system comes from measurements by the Voyager 2 mission and remote observations from Earth, and these are not sufficient to determine the presence or absence of subsurface oceans.

The large, natural moons of Uranus appear to have engaged in orbital resonances that resulted in their current orbital arrangement (Ćuk *et al.,* 2020). Extensive faulting on the surface of Miranda suggests past heating events akin to present-day heating of Europa (Beddingfield *et al.,* 2015). Evidence for relict tidal heating is also seen in Miranda's 1-billion-year-old coronae features (Plescia, 1988; Beddingfield and Cartwright, 2020). Ammonia detected on the surface of Ariel is indicative of recent geological activity because this species is expected to be destroyed on short timescales (Cartwright *et al.,* 2020). Even a small amount of ammonia mixed into their oceans may allow Titania and Oberon to retain liquid layers up to the present (Hussmann *et al.,* 2006).

Models assuming simple primordial compositions of silicates and water predict that an ocean in Miranda would have frozen out by the present era (Hussmann *et al.,* 2006). However, the presence of volatile clathrates or other insulating materials could support the persistence of liquids by providing thermal insulation and adding rigidity to the ice, both of which inhibit solid-state convection and therefore heat loss (Croft, 1987; Castillo-Rogez *et al.,* 2019; Kamata *et al.,* 2019). Evidence indicates that Saturn's moon Titan and the dwarf planets Ceres and Pluto each have methane clathrates that are expected to impede solid-state convection in ice, leading to inefficient cooling that slows the freeze-out of subsurface oceans. The presence of ammonia on the surface of Ariel suggests a similar role for volatiles that may aid in retaining heat (Cartwright *et al.,* 2020).

Uranus applies strong magnetic oscillations to its inner moons (Arridge and Eggington, 2021; Weiss *et al.,* 2021). If oceans persist within the moons Miranda, Ariel, or Umbriel, future missions may be able to use magnetic signals to detect or even characterize these oceans (Cochrane *et al.,* 2021). The larger, outer moons Titania and Oberon are subjected to weaker magnetic oscillations and spend significant portions of their orbits outside the uranian magnetosphere (Paty *et al.,* 2020). Magnetic signals are not expected to be as useful for probing for oceans at the outer moons. The potential for subsurface oceans in these moons suggests that habitability in planetary systems may extend far beyond the habitable zone. However, a more complete understanding of the prospects for life on these bodies awaits future exploration.

7.2.7.3. Triton. Triton is the largest moon of Neptune and bears a number of intriguing characteristics relating to habitability. Neptune is the farthest planet from the Sun, so Triton receives little warming in the form of sunlight. However, imagery from the Voyager 2 mission shows a mysterious ''cantaloupe terrain'' that implies a possibly very young surface (Schenk and Zahnle, 2007). Dark streaks scattered across the south polar region have been identified as plumes of material erupting from the surface (see Fig. 7.S6; Soderblom *et al.,* 1990). The source of the heating causing the geologic activity is not known, but surface features show evidence of active and extensive resurfacing (Stern and McKinnon, 2000) that may be the result of substantial warming that could sustain an ocean.

Triton orbits in a retrograde direction, unlike other large moons. The unusual orbital configuration of Triton and the arrangement of other satellites of Neptune indicates that Triton likely formed in the Kuiper Belt, similar to Pluto, and was subsequently captured by Neptune (*e.g.,* Agnor and Hamilton, 2006). If the orbit of Triton was highly elliptical upon capture, then tidal heating from circularization over time, along with radiogenic heating, could have sustained a liquid, subsurface ocean (Martin-Herrero *et al.,* 2018). Such an ocean could be only a few tens of kilometers below the icy surface (Ruiz, 2003) and, similar to Enceladus (Chapter 7.2.4), could be the source of the plumes observed by Voyager 2 coming from Triton, although these could also be the result of rapid vaporization of subsurface ices (Smith *et al.,* 1989). Few constraints exist for the composition and structure of both the surface and subsurface on Triton. Thus, future missions capable of measuring the plumes and the bulk composition of Triton are necessary to constrain the presence of a subsurface ocean and potential habitability.

7.2.7.4. Pluto. In July 2015, NASA's New Horizons spacecraft flew past Pluto (Stern *et al.,* 2015) and showed it to be an icy body with a geologically active surface. The surface age is possibly less than 10 million years in places (Grundy *et al.,* 2016; Moore *et al.,* 2016), with CH_4, N_2, and even haze particles in the thin atmosphere (with a pressure of about 1 Pa at the surface). These haze particles may be tholins, which could also comprise the reddish material observed on the surface of Pluto (Gladstone *et al.,* 2016). Gravity measurements and detailed topographic measurements taken by New Horizons indicate that a subsurface ocean is likely present beneath the icy shell of Pluto (Nimmo *et al.,* 2016), possibly insulated by a layer of clathrate hydrates (Kamata *et al.,* 2019). Thus, Pluto may sustain a habitable subsurface environment. Similarly habitable subsurface oceans could occur in other large Kuiper Belt objects (KBOs; Bierson *et al.,* 2020), which may be abundant in the outer Solar System (Brown, 2008). If Pluto and other KBOs indeed harbor habitable subsurface oceans, then habitable environments may exist well into the outer Solar System.

7.2.8. Mercury and the Moon

Neither Mercury nor the Moon are expected to be habitable today, although there is evidence that both bodies contain ice in some locations on their surfaces. Discussion of the habitability characteristics (*e.g.,* the presence of water) for these bodies is expanded in the supplementary material.

7.2.9. Giant planets

Giant planets are typically not thought to be habitable, mainly due to strong winds that cycle material between very high and very low pressures and temperatures relative to those found at Earth's surface (Ingersoll *et al.,* 2000). Discussion of habitability for giant planets is expanded in the supplementary material.

7.2.10. Habitability of exoplanets

Characterization of exoplanets is an important frontier in the search for life beyond Earth. Over the past two decades, thousands of confirmed and candidate exoplanets have been detected, mostly from dimming of the parent star when the planet passes between Earth and the star, called a transit (see Chapter 8.4.1). Transits yield an estimate of the size and orbital period, and complementary techniques can give an estimate of the mass of a planet. Of the planets that are detected, multiple factors narrow the list of candidates that may have habitable conditions, such as the planet's size and composition. These factors can be combined to yield a single metric for the prevalence of habitable planets or Earth-like exoplanets, often referred to as η_{Earth}; these factors are described in more detail in Chapter 8.4.2.

The distribution of observed planets is bimodal, with most planets categorized either as rocky or gaseous (see Fig. 3.6). The density of smaller, rocky planets seems to be approximately fixed, consistent with a relatively fixed ratio of iron and silicates (Dressing *et al.,* 2015). Gaseous planets are very likely not habitable (Chapter 7.2.9), but smaller gaseous planets may follow common pathways as they evolve that may cause them to become habitable as they lose their gas envelope (Dorn *et al.,* 2019; Rogers and Owen, 2021). This is supported by the consistent density of rocky planets, many of which may be evaporated cores of sub-Neptunes (Luger *et al.,* 2015). Sub-Neptunes are medium-sized planets with a radius between 1.7 and 4 times that of Earth, and their greater volatile inventory suggests that they may be a common source of potentially habitable water worlds. The prevalence of ice as a planetary building block implies that water-rich super-Earths (planets of 1 to about 1.7 Earth radii) may be abundant (Raymond *et al.,* 2004; Zeng *et al.,* 2019).

While numerous planets have been detected in stellar habitable zones (see Chapter 7.1.4.1 in the supplementary material), no planet has been confirmed as inhabited or even habitable. This inability to constrain exoplanet habitability arises because current telescopes are unable to characterize the atmospheres and surfaces of small, rocky planets, which are necessary observations to constrain surface habitability. Detecting signs of habitability and possibly life on exoplanets will likely require the use of telescopes currently under construction, such as ground-based extremely large telescopes (ELTs) or future space-based telescopes such as NASA's Habitable Worlds Observatory, which is based on the Habitable Exoplanet Observatory (HabEx) and Large UV/Optical/IR Surveyor (LUVOIR) space telescope mission concepts. See Chapter 8.4 for additional details on the detection and characterization of habitable and potentially inhabited exoplanets.

There are already many planetary system candidates for future telescopes to target in the search for habitability and life beyond Earth. In particular, the red dwarf star systems Proxima Centauri (Anglada-Escudé *et al.,* 2016) and TRAPPIST-1 (Gillon *et al.,* 2017) provide promising exoplanets for studying planetary habitability. Proxima Centauri hosts the closest potentially habitable exoplanet to Earth, and TRAPPIST-1 contains multiple rocky planets in its habitable zone. Other systems with rocky exoplanets in the habitable zone exist (Kane *et al.,* 2016), and future missions will likely be necessary to constrain the habitability of such worlds.

A critical uncertainty that both Proxima Centauri and TRAPPIST-1 share stems from the larger fraction of energy being emitted at the shortest wavelengths (including X-ray and ultraviolet radiation) compared to more massive stars like the Sun (*e.g.,* France *et al.,* 2016). Smaller stars are also more likely to intermittently flare, emitting tens to hundreds of times more energy than usual (*e.g.,* Davenport *et al.,* 2016). The amount and type of high-energy radiation received by a planet has implications for the origin of life (*e.g.,* Ranjan *et al.,* 2017). Additionally, high-energy radiation can drive the escape of a planet's atmosphere (see Chapter 3.4.1.5). Given the higher fluxes received by potentially habitable planets orbiting smaller stars, those planets could lose tens of Earth oceans of water during their lifetimes (*e.g.,* Fujii *et al.,* 2017). This could limit the duration of habitable conditions on these planets or even allow initially gas-dominated planets to become habitable (*e.g.,* Luger *et al.,* 2015). Given the preponderance of known planets orbiting smaller stars (*e.g.,* Dressing and Charbonneau, 2015) like Proxima Centauri and TRAPPIST-1, understanding how stellar high-energy radiation impacts planetary habitability will be the focus of future work in bounding exoplanet habitability.

This is not the only outstanding stellar question, however, as these planets face an additional hurdle stemming from their M-class hosts. Small stars take a much longer time to settle onto the stellar main sequence, an observed relationship between stellar temperature and luminosity that most stars follow (*e.g.,* Babusiaux *et al.,* 2018). As a star contracts out of a protostellar nebula, it must shed gravitational energy until it begins fusing hydrogen into helium in the core (see Chapter 3.1.1.3). For smaller stars, this transition can take tens of millions of years, during which time any rocky planets within the conventional habitable zone would lose substantial amounts of water (*e.g.,* Luger and Barnes, 2015; Bolmont *et al.,* 2017). This means that planets could meet multiple metrics for what a habitable planet should look like, such as their mass, radius, and the amount of energy they receive, but lack water, a necessary condition for life. Despite these concerns, there are several interesting targets that could be habitable.

7.2.10.1. Proxima Centauri.

The closest star to the Sun, Proxima Centauri (commonly referred to as Proxima), is a mere 4.2 light-years away. Proxima is a cool star with an effective temperature of $\sim$3000 K, about half that of the Sun, and is roughly 12% the mass of the Sun. In 2016, from radial velocity measurements, Anglada-Escudé *et al.* (2016) reported a likely rocky planet in the habitable zone of Proxima. Following exoplanetary naming conventions, the planet was named Proxima Centauri b (also called Prox. Cen. b or Proxima b).

The potential habitability of Proxima b has been considered in detail by numerous studies (*e.g.,* Ribas *et al.,* 2016; Turbet *et al.,* 2016; Kane *et al.,* 2017; Meadows *et al.,* 2018; Del Genio *et al.,* 2019; Salazar *et al.,* 2020). From various climate and planetary models, studies of the habitability of Proxima b have found that it may indeed support a habitable surface, and possibly life. However, as is the case with all planets, our understanding of habitability on Proxima b depends on numerous factors, including atmospheric composition, possible spin–orbit resonance, precise mass and

radius, orbital eccentricity, outgassing rate, stellar activity, and more (see Chapter 7.1.4.1 in the supplementary material). The James Webb Space Telescope (JWST) may be able to constrain some of these properties, even though Proxima b does not transit its host star (Turbet *et al.,* 2016).

7.2.10.2. TRAPPIST-1. Approximately 40 light-years from Earth sits TRAPPIST-1, a cold M dwarf star with an effective temperature of just ~2600 K, less than half that of the Sun. TRAPPIST-1 hosts seven small planets named TRAPPIST-1 b, c, d, e, f, g, and h, with b the innermost and h the outermost (Gillon *et al.,* 2017). The planets span the width of the habitable zone of TRAPPIST-1 and beyond, providing a planetary system where the effect of orbital distance on habitability can be probed.

The potential habitability of the TRAPPIST-1 planets has been extensively modeled (*e.g.,* O'Malley-James *et al.,* 2017; Vida *et al.,* 2017; Wolf, 2017; Dong *et al.,* 2018; Lincowski *et al.,* 2018; Papaloizou *et al.,* 2018). Further, the TRAPPIST-1 planets transit their star, making it possible to characterize their mass and radius (*e.g.,* Grimm *et al.,* 2018). While models of the TRAPPIST-1 planets indicate they may be habitable, particularly d and e, future observations from ELTs and JWST will further constrain the potential habitability of these planets and make detailed measurements of their atmospheric composition (*e.g.,* Lincowski *et al.,* 2018).

7.2.10.3. Considerations for water worlds. Water worlds (often taken to mean > 1% H_2O by mass) are an important class of rocky exoplanets. Water worlds are expected to be quite common (Raymond *et al.,* 2004; Zeng *et al.,* 2019), and their relatively high potential for habitability (Kite and Ford, 2018) renders them a vital topic for study. For comparison, Europa is at least 6% H_2O by mass (Anderson *et al.,* 1998), and Earth has well below 1% H_2O by mass.

The composition, chemistry, and dynamics of the oceans of water worlds are critical to understanding their potential habitability. The large mass fraction of water required implies that water worlds are made mainly of cometary material (Léger *et al.,* 2004). CO_2 is a common volatile component of cometary material (Mumma and Charnley, 2011). To prevent overheating by a strong greenhouse effect, an efficient CO_2 sequestration mechanism may be important (Marounina and Rogers, 2020), perhaps implying that the habitable zone for water worlds is shifted farther from the star. Outgassing of magmas underneath an ocean may be suppressed by high overburden pressure (Journaux *et al.,* 2020; Krissansen-Totton *et al.,* 2021). This likely reduces geochemical cycling and feedbacks (Noack *et al.,* 2016), but it is not necessarily a detriment to the maintenance of habitable temperatures (Kite and Ford, 2018). The presence of high-pressure ice phases deep in the oceans of water worlds may impede water–rock reactions (Journaux *et al.,* 2020; Marounina and Rogers, 2020) thought to be important for creating habitable conditions by preventing contact between liquid water and silicates. However, convection in high-pressure ices may facilitate mixing and contact between the ocean and silicates (Kalousová *et al.,* 2018).

Because sub-Neptunes make up a large fraction of the exoplanets detected to date (Otegi *et al.,* 2020), understanding whether and how they may evolve into water-rich super-Earths is essential to assessing the potential habitability of such planets. Water-rich sub-Neptunes may lose their volatile envelope as they migrate close to their parent star or as the star brightens over time, leaving behind a water world (Luger *et al.,* 2015). Dissolution of H_2O into iron-rich silicates may buffer the water content of sub-Neptunes as they lose the remainder of their volatile envelope, potentially maintaining a long-lived, H_2O-dominated atmosphere (Kite and Shaefer, 2021) that may overlay a water world. Ultimately multiple, common planetary evolution pathways may result in the formation of water worlds.

7.3. Conclusion

Understanding the extant conditions and history of the bodies in the Solar System, especially in the context of life as we know it on Earth, is vital to the study of how life gains a foothold and evolves along with planetary bodies. Applying methods from a wide variety of scientific disciplines is an essential part of our developing knowledge of habitable environments in the Solar System. The more we learn about Earth and the potentially habitable worlds in our planetary neighborhood, the better we are able to apply those lessons to other star systems and the multitudes of worlds they contain. The same is true in reverse—studying the plethora of known exoplanets puts the Solar System into context.

Assessing the habitability of a particular environment is an exercise in synthesis. The study of habitable worlds is at its most valuable when coupled with exploration, in the form of searching for signs of life, past and present. The next chapter in this primer, Chapter 8, will discuss biosignatures—what signals we may look for in different environments and what these signals may tell us about possible inhabitants.

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All authors provided editing feedback for the entire manuscript.

Z.S. Cooper authored sections on biological requirements for habitability and co-authored sections on chemical requirements for habitability and scales of habitability.

D.M. Glaser authored sections on stellar and planetary requirements for habitability and co-authored the section on scales of habitability.

O. Lehmer authored sections on Enceladus and Titan and co-authored sections on other icy worlds and exoplanets.

V. Mierzejewski authored the section on chemical requirements for life.

M.J. Styczinski managed the team and authored the abstract, section introductions, and sections on Venus, Europa, Ganymede, and Callisto, and giant planets, and coauthored sections on Enceladus, Mars, exoplanets, stellar and planetary requirements for habitability, and scales of habitability.

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Supplementary Material

Supplementary Information
Supplementary Figure 7.S1
Supplementary Figure 7.S2
Supplementary Figure 7.S3
Supplementary Figure 7.S4
Supplementary Figure 7.S5
Supplementary Figure 7.S6

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Abbreviations Used

ELTs = extremely large telescopes
JWST = James Webb Space Telescope
KBOs = Kuiper Belt objects
RSL = recurring slope lineae